Fake tales of bodymass models

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Abstract: TODO.

Keywords: food webs

- Body-size-based allometric trophic network (ATN) (Brose et al., 2006) models predict interactions from consumer–resource size relationships.
- However, body size is only one determinant of trophic interactions; other traits and ecological constraints may be important.
- **Aim:** quantify how often body-size-based ATN models predict interactions absent from empirical metawebs, and test whether network structure can nevertheless be recovered.
- Prediction: ATN models will generate a high proportion of forbidden interactions, even when their broad structural properties resemble those of empirical metawebs.

Template metawebs

- **Great Britain:** European metaweb (Maiorano et al., 2020); subsampled to mammal species checklist from the UK Mammal Society - this is because some of the links in the european metaweb are 'forbidden' due to co-occurrence limitations but could be feasible if species are treated as co-occurring. So this would have inflated the number of perceived forbidden links in the ATN. For Canada it is less of a problem since it is not constrained by co-occurrence.
- **Canada:** probabilistic metaweb from Strydom et al. (2022).
- **Body sizes:** mammalian body-mass estimates from PanTHERIA (Jones et al., 2009).
- Briefly describe species matching, taxonomic harmonisation and missing-data handling.

ATN models and forbidden interactions

- Generate probabilistic and binary ATN networks using body size, varying $(R_{\text{opt}}), (\gamma)$ and, for binary networks, the interaction threshold.
- For probabilistic metawebs, account for uncertainty by sampling interactions.
- Define forbidden interactions as ATN links absent from the corresponding metaweb.
- Quantify the percentage of ATN links that are forbidden across parameter combinations and simulation replicates.

Network structure comparison

- Compare a small set of structural metrics between ATN networks and metawebs (e.g. connectance, degree distributions and nestedness, where appropriate).

- Assess whether similar structural properties can coexist with poor recovery of specific species-level interactions.

Body-size models predict many forbidden interactions

- Report the percentage of predicted ATN links absent from the metaweb.
- Show sensitivity to $(R_{\text{opt}}), (\gamma)$, and the binary interaction threshold.
- Compare results between Great Britain and Canada.

Similar structure does not imply correct interactions

- Compare structural metrics between ATN networks and metawebs.
- Highlight any cases where structural similarity coexists with low pairwise interaction agreement.

Discussion

- Body-size-based ATN models can predict interactions that are unsupported by the reference metawebs.
- Structural similarity, if observed, does not necessarily imply recovery of the correct species-level links.
- Discuss the limitations of body size as a sole predictor and the potential importance of additional traits and ecological constraints.
- Note that metaweb absence is not definitive proof that an interaction is biologically impossible; results measure disagreement with the reference interaction data.

Data and code availability

- Provide data sources, analysis code and processed outputs.

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